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Listening in a Noisy Environment



HAVE YOU EVER FACED A SITUATION WHERE YOU REALLY WANTED TO ENJOY your favorite music in a very noisy environment? To overcome this problem, you can either block the noise or increase the music level to mask the annoying sounds. This begets the question, “Is there a better solution?” In recent years, the technology industry has grown enormously, especially in the areas of smart, mobile, and wearable electronic products. Many companies are trying to gain consumers’ attention by developing high-quality

*Integration of
active noise control
in audio products.*

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products equipped with advanced noise reduction. Among them, several companies have successfully used active noise control (ANC) techniques to cope with noise problems.

ANC systems generate antinoise to cancel undesired sounds based on the superposition principle [1]–[3]. In audio applications, audio quality is a priority, which requires ANC systems to cancel the noise without degrading the desired audio signals. In addition, we have to avoid any scenario in which the audio signal interferes with the ANC system. Fortunately, the advanced development of the audio-integrated

ANC algorithm makes that possible [4], [5]. This article explores the technologies and applications of ANC integrated with audio products.

AUDIO-INTEGRATED ANC TECHNIQUES

ANC utilizes secondary audio sources to generate antinoise of equal amplitude and opposite phase (180° out of phase) that cancels the undesired primary noise, resulting in quiet residual noise. In practice, the generation of antinoise is usually done digitally and adaptively to achieve highly precise control, temporal stability, and reliability [1]. The ANC design usually utilizes an appropriate array of microphones as sensors and electronically driven loudspeakers as the secondary sources. The analog components—loudspeakers, mixers, amplifiers, and microphones—used in ANC systems are already available in audio products. This allows the integration of both audio and ANC systems into a single audio-integrated system sharing analog components. This integrated system, developed in 1993, applies the audio interference cancellation algorithm [4] to allow both ANC and audio functions to work simultaneously without interfering with each other. That is, ANC attenuates only unwanted noise without canceling the desired audio.

In general, there are two basic types of audio-integrated ANC algorithms: feedforward and feedback [1]. Feedforward ANC utilizes a reference microphone at the upstream to provide a reference signal of the approaching noise. Primary noise that correlates with the reference signal will be canceled downstream at the error microphone. In a feedback ANC system, the reference signal is internally generated as a predicted signal based on the measured error signal. Therefore, the feedback ANC system can only reduce predictable noise, while the feedforward ANC system is capa-

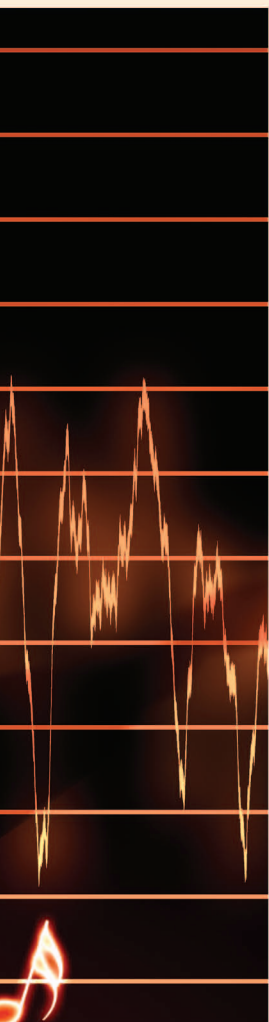
ble of reducing broad-band noise. In ANC applications such as headphones, the secondary loudspeaker of the ANC system is also used to play intended audio signals (such as speech and music) during ANC operation. To prevent the ANC system from canceling the desired audio signal and to avoid the audio signal acting as interference that degrades the ANC performance, the audio-integrated ANC algorithm was proposed.

FEEDFORWARD AUDIO-INTEGRATED ANC SYSTEMS

The feedforward ANC system is exemplified by canceling noise in a duct, as illustrated in Figure 1. The primary noise is produced by the noise source, and a reference microphone is placed upstream for sensing the primary noise before it reaches the downstream secondary loudspeaker in the duct. The reference signal $x(n)$, which is obtained through a preamplifier, an anti-aliasing filter, and an analog-to-digital converter (ADC), is processed by the adaptive filter $W(z)$ to generate the antinoise signal $y(n)$ that drives a secondary loudspeaker, through a digital-to-analog converter (DAC), a reconstruction filter, and a power amplifier. At the downstream, the error microphone senses the residual noise $e(n)$, which is obtained through a preamplifier, an anti-aliasing filter, and an ADC. The adaptive filter $W(z)$ minimizes the error signal $e(n)$ by adapting filter coefficients automatically using the filtered-X least mean square (FXLMS) algorithm [6]. The secondary-path transfer function $S(z)$ is from $y(n)$ to $e(n)$, which includes the DAC, reconstruction filter, power amplifier, loudspeaker, acoustic path from the loudspeaker to the error microphone, preamplifier, anti-aliasing filter, and ADC. The secondary-path model $\hat{S}(z)$, which can be estimated by either the offline [1], [7] or online [8] method, is required by the FXLMS algorithm to compensate for the effects of the secondary-path transfer function $S(z)$.

Based on the feedforward ANC system, the feedforward audio-integrated ANC system is shown in Figure 2, where the audio signal $a(n)$ is mixed with the adaptive filter output $y(n)$, and the composite signal $u(n)$ is used to drive the secondary loudspeaker. Thus, the signal $e(n)$ picked up by the error microphone contains both the residual noise and the audio component. The audio source $a(n)$ is filtered by the secondary-path estimation filter $\hat{S}(z)$ to estimate the audio component picked up by the error microphone, and then the estimated audio component $a'(n)$ is subtracted from $e(n)$ to obtain the audio-free error signal $e'(n)$, which is used to update the adaptive filter $W(z)$. Therefore, the adaptive filter $S'(z)$ performs the adaptive canceling of the audio component picked up by the error microphone using the least mean square (LMS) algorithm [9].

As shown in Figure 2, the adaptive filter $\hat{S}(z)$ performs the adaptive system identification of the secondary path $S(z)$ using the audio signal $a(n)$ as the excitation signal. In theory, when the audio signal is rich in frequency content and uncorrelated with the antinoise $y(n)$, a perfect model can be obtained, i.e., $\hat{S}(z) \cong S(z)$. Thus, the error signal $e'(n)$ used for the FXLMS algorithm is the true residual noise without audio component. Therefore, the performance of the FXLMS



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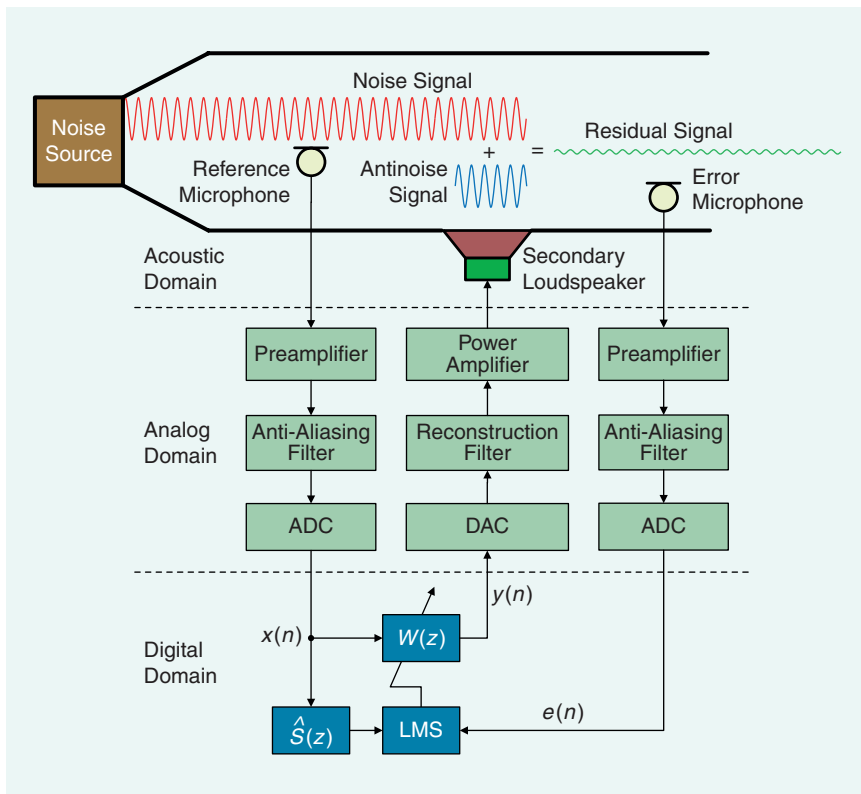


FIGURE 1. A block diagram of the feedforward ANC system.

algorithm will not be degraded by the additional audio component picked up by the error microphone, and the ANC system will not cancel the desired audio component, because the audio component is not fed back to the FXLMS algorithm. The additional benefit of using the audio-integrated algorithm is that the adaptive filter $\hat{S}(z)$ performs the online modeling of the secondary path using the audio signal as an excitation signal. Furthermore, the audio-integrated ANC system saves on overall system cost by using the same analog components—loudspeakers and amplifiers to play the intended audio signal.

FEEDBACK AUDIO-INTEGRATED ANC SYSTEMS

The concept of a single-channel feedback ANC system is illustrated in Figure 3 for the duct noise problem. The FXLMS algorithm, the secondary path $S(z)$, and its estimation $\hat{S}(z)$ are the same as in the feedforward ANC system.

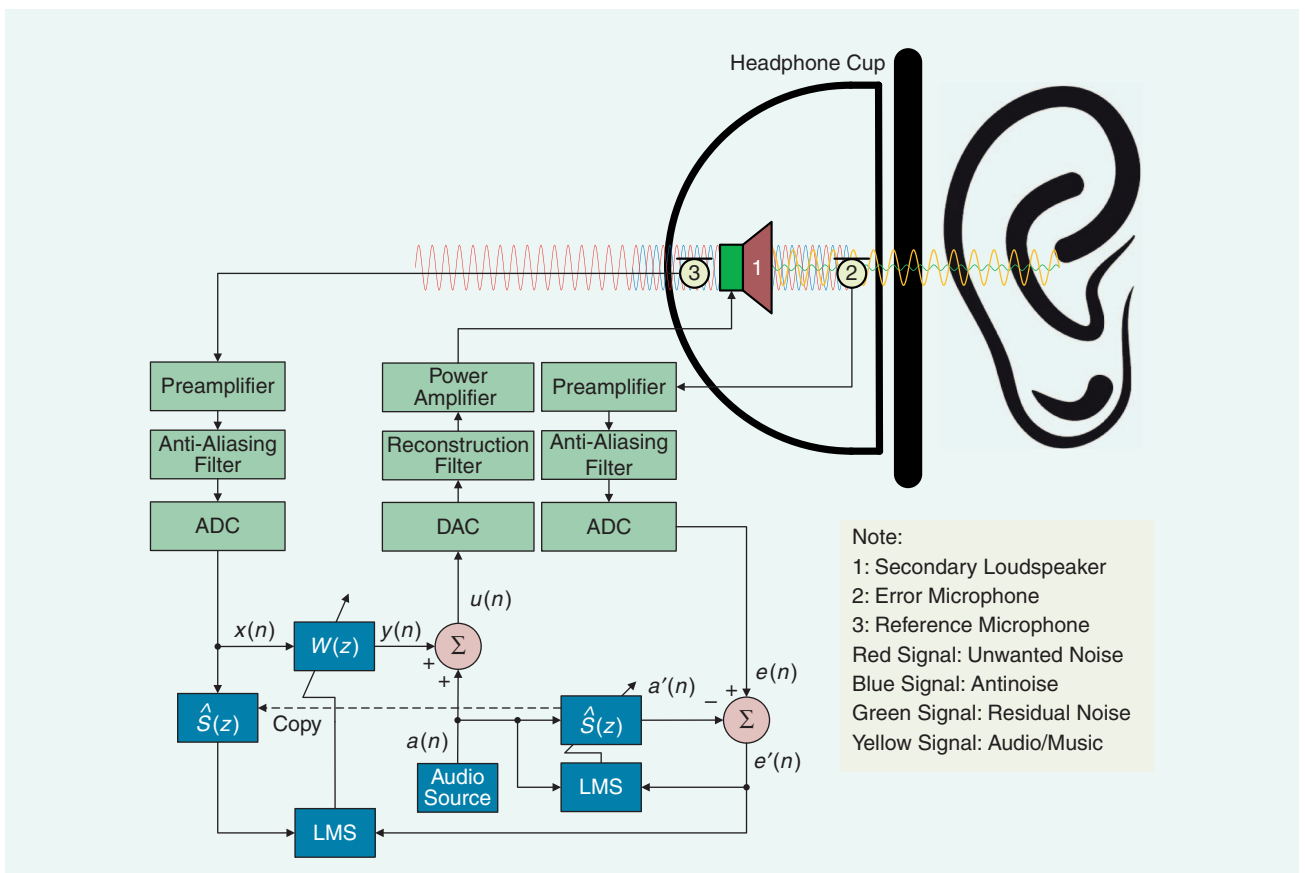


FIGURE 2. A block diagram of the audio-integrated feedforward ANC system.

There is no reference microphone, so the reference signal $x(n)$ is synthesized as $x(n) \cong \hat{d}(n) = e(n) - y'(n)$, where $y'(n)$ is the adaptive filter output $y(n)$ filtered by the secondary-path model $\hat{S}(z)$. As shown in Figure 3, the predicted noise $x(n)$ will be equal to the primary noise if $\hat{S}(z) \cong S(z)$. Because the current estimate signal $x(n)$ is used as the reference signal for the next iteration, i.e., at time $(n + 1)$, the reference signal synthesis process is functioned as a one-step predictor. This principle indicates that the adaptive feedback ANC can cancel only predictable noise. Figure 4 shows the audio-integrated feedback ANC system. Its principle is similar to the feedforward ANC structure described in Figure 2.

PRACTICAL APPLICATIONS IN CONSUMER ELECTRONICS AND HEALTH CARE

This section introduces several practical applications of audio-integrated ANC developed in the Active Noise Control Laboratory at Chung Yuan Christian University, Taiwan. There are more than ten ANC stations in the lab; only those used for audio-integrated ANC applications are illustrated in this section.

ANC CLIP HEADPHONES

Wearable listening devices are very popular nowadays. People tend to enjoy music delivered from electronic devices such as mobile phones. An ANC headphone applies audio-integrated ANC techniques to simultaneously cancel the ambient noise and deliver high-quality sound [10]–[14]. Currently, several ANC headphones are available for use in everyday life. In fact, the ANC headphone is the most commercially successful audio-integrated ANC product.

There are many types of headphones, such as on-ear, overear, in-ear canal, earbuds, canalbuds, and clip headphones. The majority of commercial ANC headphones are of the on-ear and overear varieties. They are large in size, heavy, uncomfortable to wear, and cumbersome to carry, but they offer the potential of maximum bass and loudness as well as passive blocking of high-frequency noise. In-ear canal, earbuds, and canalbuds headphones are lightweight and compact, and offer good noise isolation, but their sound quality is poorer. In between these two groups is the clip headphone, which offers light weight, compactness, and adequate sound quality. However, it has poor passive noise isolation, which provides the main challenge for noise reduction. Therefore, an ANC clip headphone has been developed in this lab.

The ANC clip headphone (Figure 5) uses the feedback ANC algorithm,

which is used in most ANC headphones. The error microphone is placed inside the earpad. The clip headphone's speaker is placed inside an enclosure made by a three-dimensional printer.

ANC HEADRESTS

Sitting on a chair may be the most frequent activity in a human being's daily life. Unfortunately, when one is sitting in a noisy environment, such as an aircraft cabin, one's hearing may be damaged if the noise level is above 80 dB. So passengers have the option of wearing ANC headphones to protect their hearing. However, wearing such a device for long periods is uncomfortable. Therefore, a comfortable ANC headrest device may be used to offer noise cancellation. Unlike the noise-canceling headphone, which attaches to the ear and cancels the noise in an enclosed space, the ANC headrest system cancels the noise in open space, which has potential applications in aircraft, trains, buses—in fact, any place where there are seats in a noisy environment [15]–[19].

The basic idea of the ANC headrest is to establish a quiet zone around the user's ears. In general, the quiet zone is created for covering both ears by utilizing two secondary loudspeakers and two error microphones located close to the ears, as shown in Figure 6. The secondary loudspeakers generate the antinoise to cancel the noise at the error microphones. This technique is known as *local noise cancellation*, in which

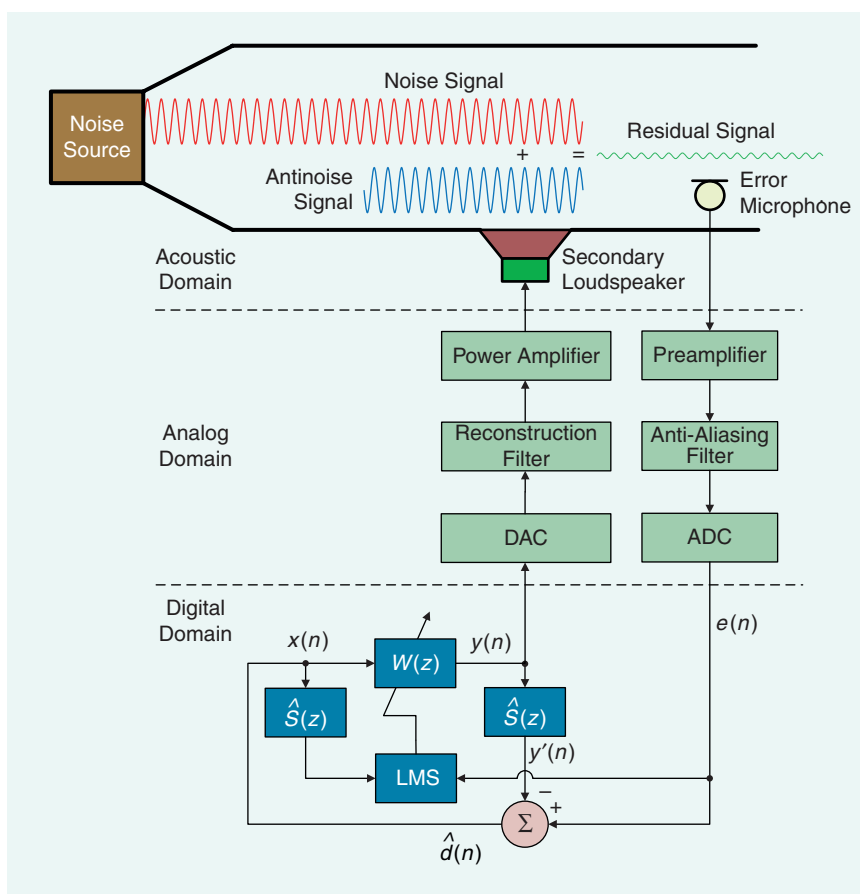


FIGURE 3. A block diagram of the feedback ANC system.

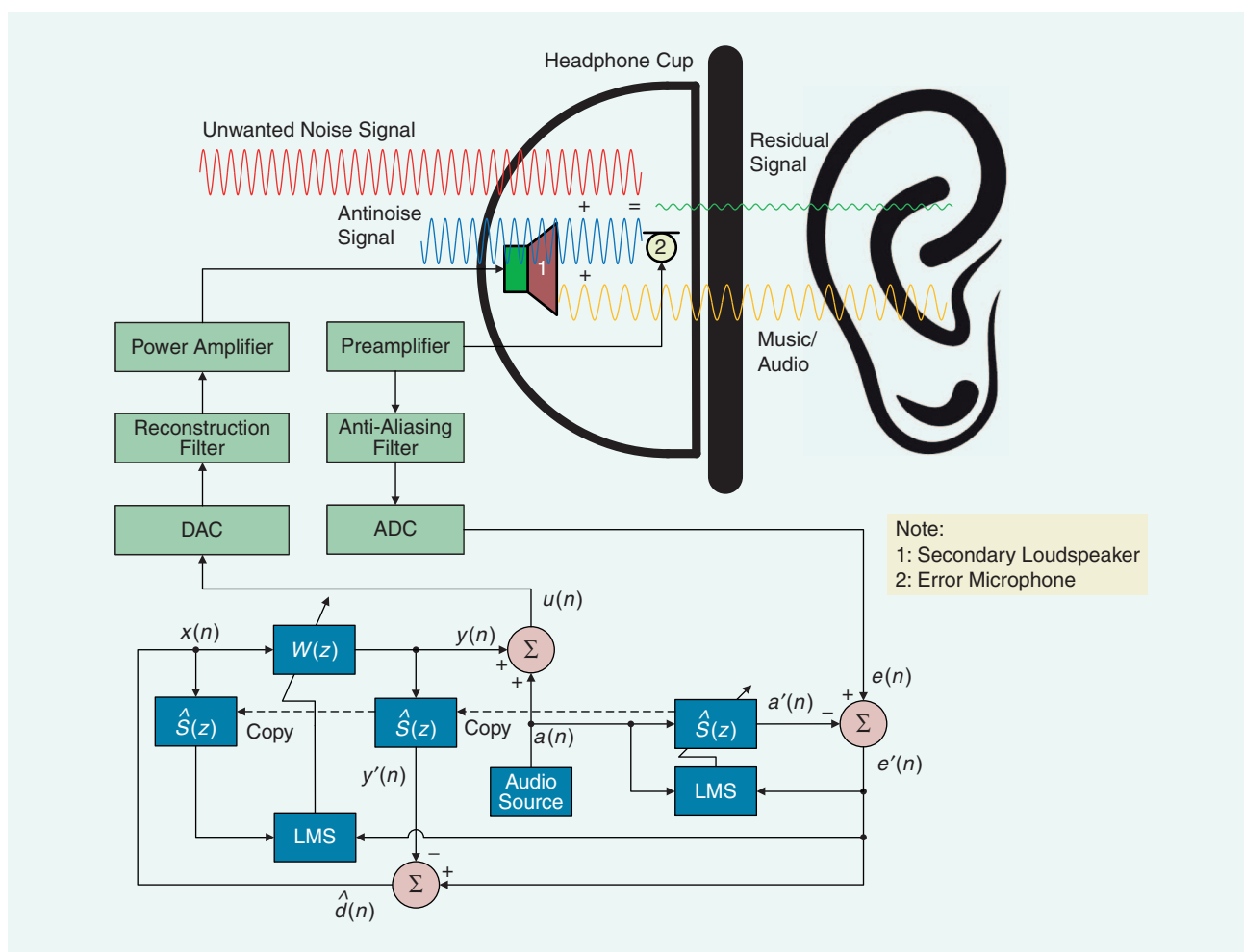


FIGURE 4. A block diagram of the audio-integrated feedback ANC system.

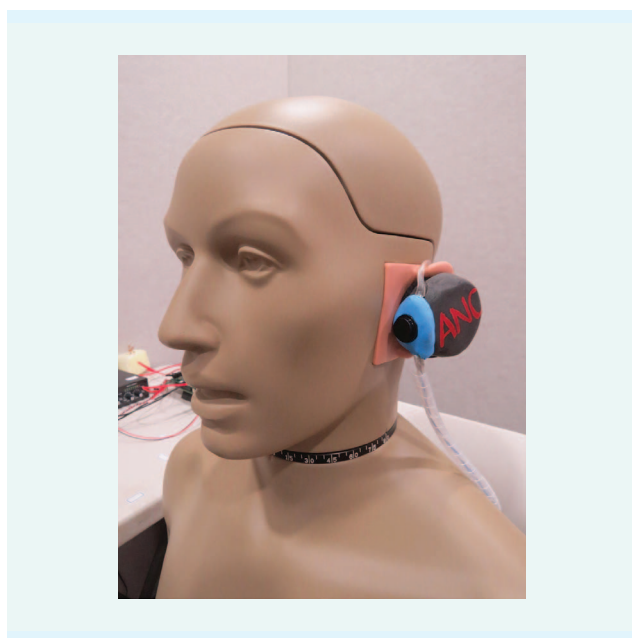


FIGURE 5. An ANC clip headphone with KEMAR.

noise cancellation is achieved only at the center of the quiet zone close to the head.

Stereo music, as well as hands-free communication, can be integrated as additional functions. The secondary loudspeakers play stereo music and antinoise simultaneously. The error microphones can capture the voice of the person who sits on the chair when hands-free communication is activated. The stereo music can also provide residual noise masking.

ANC PILLOWS

The health-care industry is one of the world's largest and fastest-growing industries. In 2011, US\$6.9 trillion was spent on health worldwide, and each person on average spent US\$1,008 per year. In 2013, health care consumed 17.1% of the U.S. gross domestic product. Motivated by these facts, we developed an innovative ANC pillow for health-care applications [20]–[22].

An ANC pillow developed in this lab provides several health-care functions, including noise protection, nature sound therapy, apnea and sleep disorder detection, and sleep quality monitoring. The noise protection is achieved

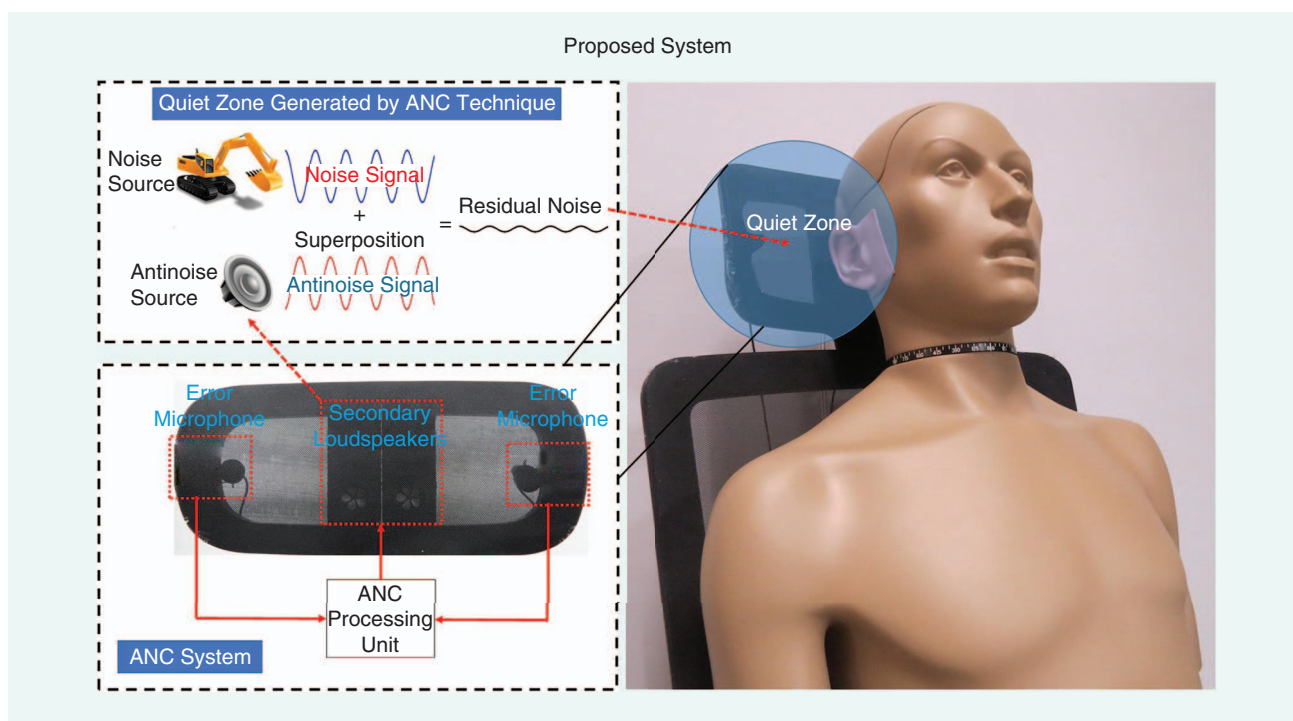


FIGURE 6. An ANC headrest system.

by the ANC system using two secondary loudspeakers and two error microphones installed inside the pillow, as shown in Figure 7. The goal of applying ANC is to reduce snore noise from one's partner in bed. The reference microphone is placed close to the bed partner to enable the feedforward ANC algorithm. Similar to the ANC headrest, a quiet zone is created to cover both ears. The main challenge for the snore ANC is that a snore is characterized by fast-changing and intermittent noise. This requires fast convergence adaptive algorithms.

The same secondary loudspeakers can also play nature sounds for residual noise masking as well as psychoemotional relaxation. This integrated system is achieved by applying audio-integrated ANC techniques. In addition, pressure sensors are installed to provide sleeping disorder and quality measurements. All these functions can be processed using a single digital signal processor.

ANC HEARING AIDS

According to a March 2015 World Health Organization (WHO) report, over 5% of the world's population—or some 360 million people—experience hearing loss [23]. Therefore, the need for assistive devices such as hearing aids is essential to improve their hearing. Hearing aids are designed to amplify sound for the user to correct impaired hearing, usually with the aim of making speech more intelligible.

There are two major types of hearing aids: in-the-ear and behind-the-ear. The behind-the-ear type can be further separated into two different varieties based on how they send the

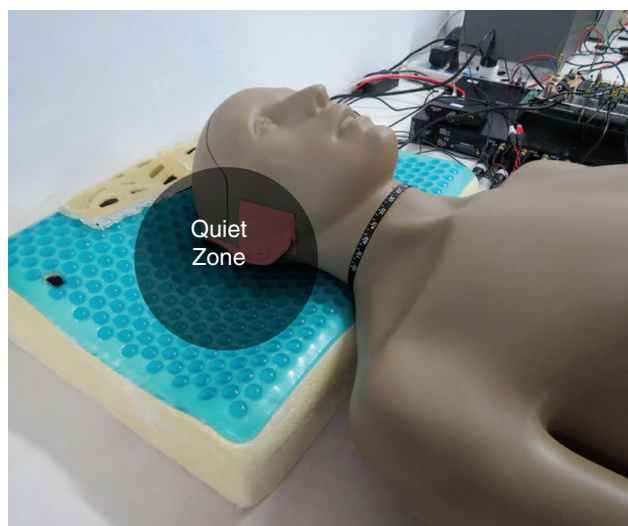


FIGURE 7. An electronic pillow.

amplified signal into the ear canal. The first is called *closed fit* (or *closed air*), which can isolate undesired noise due to a fully blocked canal, as shown in Figure 8(a). Users of this type of device, however, might feel uncomfortable after long wearing. The other type of hearing aid, known as *open fit* (or *open air*), is able to solve this problem by replacing the earphone with a thin tube that stretches into the ear canal, which allows normal air flow. Unfortunately, noise leakage becomes an issue due to the unblocked ear canal [Figure 8(b)]. This undesired noise propagates directly to the eardrum, cannot be canceled by the noise reduction

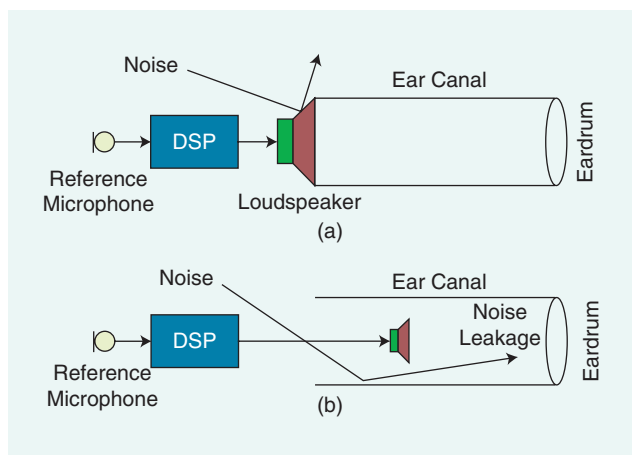


FIGURE 8. An illustration of the noise effect in different types of hearing aids: (a) closed fit and (b) open fit.

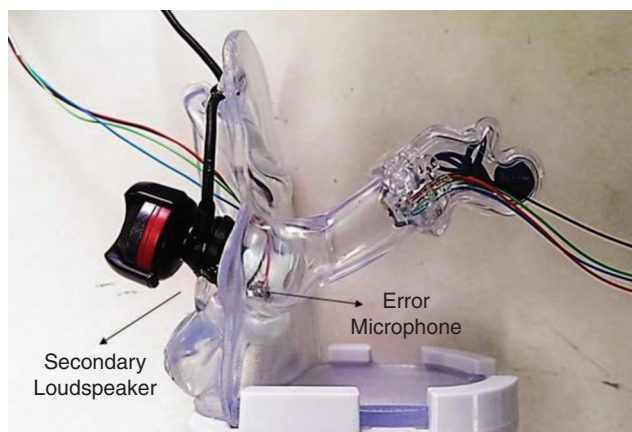


FIGURE 9. An experimental setup of an audio-integrated ANC for an open-fit hearing aid.

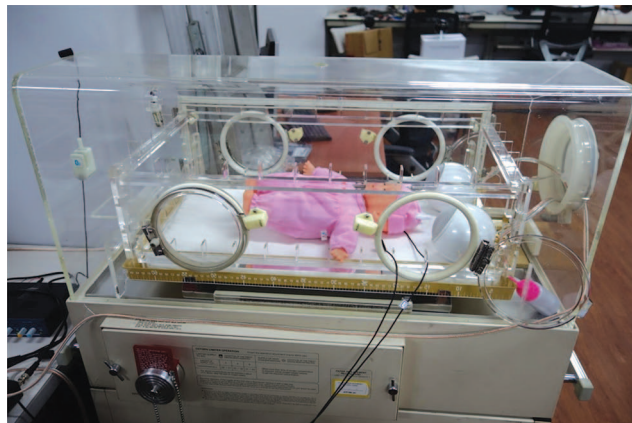


FIGURE 10. An audio-integrated ANC system for an infant incubator.

function of digital hearing aids, and interferes with speech. To solve the noise leakage problem, audio-integrated ANC techniques can allow ANC and speech functions to work simultaneously [24]–[27].

For simplicity of development, we use an earphone as a hearing aid in the lab. An additional error microphone is installed on

the earphone close to the eardrum (Figure 9). The secondary loudspeaker is shared with the existing loudspeaker of the earphone. To perform both the ANC function and the hearing aid functions, the audio-integrated feedback ANC algorithm is required.

ANC INFANT INCUBATORS

Every year, about 15 million babies are born preterm (before 37 completed weeks of gestation), and this number is rising [28]. The use of infant incubators in neonatal intensive care units (NICUs) significantly helps such premature and medically challenged infants. The infant incubator is an enclosure with sections that are transparent for viewing the baby within. Some sensors and devices are placed inside the incubator for monitoring the baby's vital statistics and for controlling the temperature, humidity, supplementary oxygen, and other parameters.

High levels of noise from these internal devices, from medical equipment outside the incubator and human activities inside the NICU, can result in numerous adverse health effects for an infant, including sleep disturbance and other forms of stress. The internal ear structures of a baby may not yet be fully developed, so an infant can develop significant long-term hearing loss.

Passive noise reduction is not effective for low-frequency noise. Furthermore, most passive techniques either take up limited incubator space or block the view of an infant to caregivers, or both. These difficulties motivated the development of ANC systems to cancel low-frequency noise inside incubators [29]–[32].

The goal of applying ANC is to establish a quiet zone around the infant's ear inside the incubator. Therefore, two secondary loudspeakers are installed inside the incubator along with two error microphones, as shown in Figure 10. A challenging problem for an incubator ANC is that noise comes from multiple noise sources inside and outside the incubator.

In addition to noise cancellation, therapeutic functions are essential for enhancing treatment and optimizing premature infants' physical and neurological development. A parent's voice has been shown to enhance vocalization in premature infants [33]. Music and sound therapy, such as womb sounds that provide infants with the ambience of the mother's womb, have also been shown to benefit infants in such ways as weight gain, reduced observed stress behaviors, shorter term of hospitalization, and increased oxygen saturation levels for short periods of time [34]. Therefore, the incubator ANC also can apply audio integration techniques to allow the secondary loudspeakers to play the sound without interfering with the ANC function. Besides, crying detection or other sound monitoring can be added, since the error microphones are available for use.

ANC HELMETS

Motorcycle noise levels on an open road are in the range of 78–90 dBA at 30 mi/h to 116 dBA at 120 mi/h, depending on the type of helmet, motorcycle, and road condition [35]. According to the WHO report, continuous exposure to noise levels of 95 dBA for more than 47 min may result in various hearing impairments, such as hearing loss, tinnitus, or acoustic shock, and may even lead to accidents [36].

Unfortunately, commercial helmets provide only physical protection, not noise protection, especially from noise at low frequencies. In general, a motorcycle helmet's liner can act as a passive barrier to provide attenuation of noise (such as the wind noise) at frequencies above 500 Hz. However, the dominant motorcycle noise is at low frequencies. Therefore, combining ANC with the helmet liner's passive noise control can provide broad-band noise control for motorcycle riders [37]–[40]. Basically, the design of ANC helmet systems is similar to that of ANC headphones. Therefore, as with ANC headphones, the secondary loudspeakers and error microphones are installed inside the earpad close to the ear, as shown in Figure 11. Since motorcycle noises are narrow-band noises containing several harmonics, the feedback ANC system is also effective.

In addition, an audio integration system can be implemented. An audio/speech listening capability is added to allow motorcycle riders to listen to music/speech from audio devices using the same loudspeakers. A reference microphone can be installed inside the helmet near the mouth location. This arrangement enables the system to apply the feedforward ANC algorithm, which helps to cancel noise and pick up the voice of the rider for communication.

EXPERIMENTAL RESULTS

This section examines the real-time performance of the audio-integrated ANC applications introduced above. All of the experiments were verified by practical experimental setups using a Texas Instruments (TI-TMS320C6713) DSP Starter Kit (DSK). It is important to note that for some applications, low-cost and low-power-consumption microcontrollers that satisfy computational requirements could also be used in real-world products.

The first experiment shows the performance of tonal noise cancellation for the ANC clip headphone shown in Figure 5. When tonal noise was played at different frequencies (400–3,000 Hz, in steps of 100 Hz), the noise reduction (the noise level difference between ANC off and ANC on) was measured at the microphone inside the KEMAR ear, as shown in Table 1.

For the ANC headrest system, two speakers mounted inside the headrest were used to drive the antinoise. Two omnidirectional microphones were used as the error microphones. The performance of the audio-integrated ANC head-

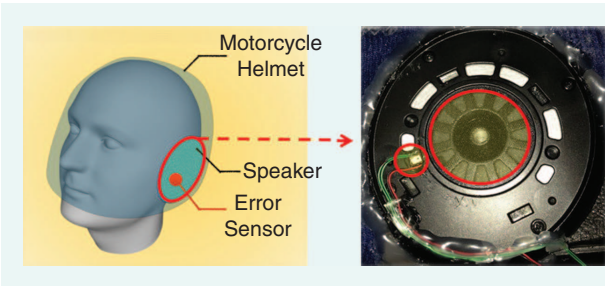


FIGURE 11. The configuration of a speaker and microphone inside a helmet (left ear).

rest is shown in Figure 12, where the black dashed line is the result of playing the audio signal using the secondary speakers and measuring by the error microphones. The blue line shows the audio signal with the noise, and the red line presents the performance of ANC with audio integration. This clearly shows that the red line is almost identical to the black line, meaning that the noise can be reduced by the ANC system and that the residual signal is close to the musical signal, without noticeable degradation.

Figure 13 shows the performance of the ANC hearing aid integrated with a speech signal. The red color in the spectrogram plot represents high power and the green is for low power. We used three tonal signals at 1,500 Hz, 1,800 Hz,

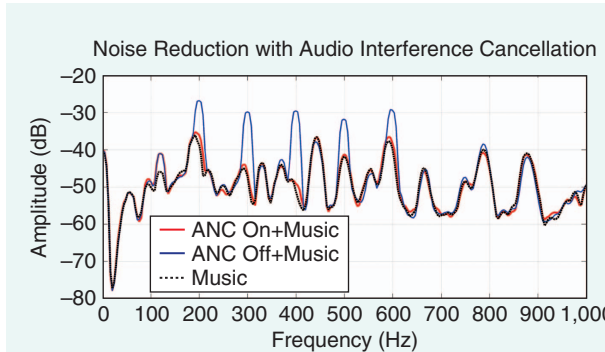


FIGURE 12. The performance of an ANC headrest system with audio integration.

Table 1. The tonal noise reduction achieved by an ANC clip headphone at different frequencies.														
Frequency (kHz)	0.4	0.5	0.6	0.7	0.8	0.9	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7
Noise Reduction (dB)	24.3	25.4	23.7	24.4	23	15.5	20.9	18.5	28.6	25.2	21.3	20.4	21.3	28.9
Frequency (kHz)	1.8	1.9	2	2.1	2.2	2.3	2.4	2.5	2.6	2.7	2.8	2.9	3	
Noise Reduction (dB)	25.8	20.4	22.5	16.5	18.2	19.1	19.3	18.6	19.7	17.5	19.5	23.5	19.6	

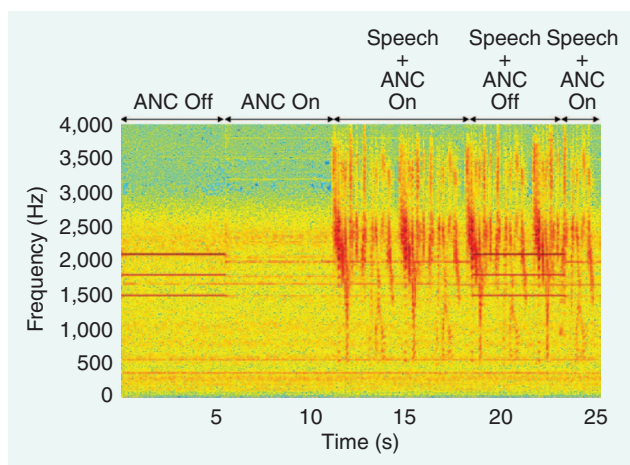


FIGURE 13. An audio-integrated ANC hearing aid.

and 2,100 Hz as the noise to disturb the speech signal. The spectrogram plot clearly shows the tonal noise with high power in the first 6 s before the ANC function turned on. After that, we integrated the tonal signals with an audio signal and showed that the ANC system still performed well in a situation of speech integration.

Figure 14 illustrates the performance of an ANC helmet. We attached a microelectromechanical systems microphone on the side of the speaker at each ear (see Figure 11), and the result was obtained from the microphone. First, we played the recorded motorcycle noise using another speaker as the primary noise and then played the audio signal at about 21 s to examine the performance of listening to music in a noisy environment. The power of motorcycle noise is concentrated in a low-frequency range. When the ANC function was turned on at about 9 s, most noise was reduced. Also, the audio integration performance shown in Figure 14 verifies the effectiveness of the audio-integrated ANC system.

There are some challenges in practical applications of audio-integrated ANC systems, including the secondary-path

modeling error and the multiple sampling rates. In addition, for those ANC applications where quiet zones are created in open space, the size and location of a quiet zone should appropriately cover both ears. Furthermore, the location of noise sources may be unknown; therefore, the adaptive algorithm has to be effective to track and synthesize reference noise without using reference microphones.

CONCLUSIONS

ANC offers an effective, low-cost, lightweight, and compact solution to noise problems in audio and health-care products. As these products are used in noisy environments, audio-integrated ANC systems are essential for enhancing the audio quality and canceling annoying noise. In this article, audio-integrated ANC algorithms were briefly explained and several potential audio-integrated ANC products were discussed. Real-time experiments were conducted to verify the performance of noise reduction. We hope this article will motivate the development of more and more practical ANC applications in audio products in the future.

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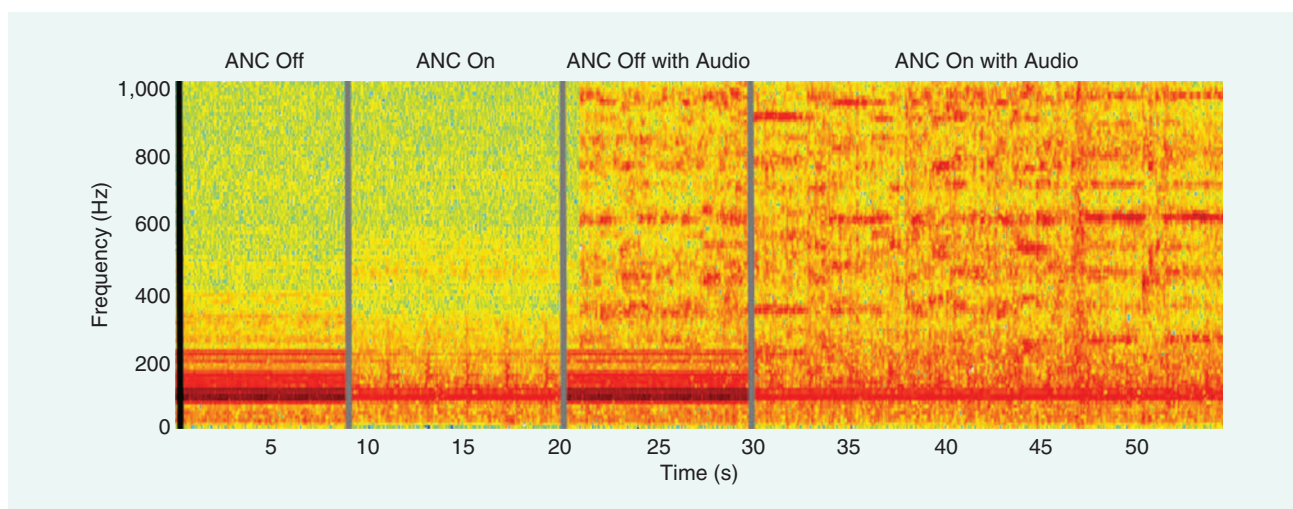


FIGURE 14. The performance of the ANC helmet with audio integration.

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